

E-COMMERCE PERFORMANCE ANALYSIS

Identifying sales performance, product contribution & refund behaviour

OBJECTIVE:

The objective of this analysis is to evaluate overall e-commerce performance by analysing sales, revenue contribution, pricing and refund behaviour across products. The goal is to identify key revenue drivers, high performing products, and potential area of revenue leakage and to provide actionable business insights using a combination of SQL, Python & Power BI.

DATASET OVERVIEW:

1. Data source: Maven Analytics
2. Key tables used: Website Sessions, Orders, Order Items, Products, Refunds.

DATA CLEANING & PREPARATION:

The dataset was initially available in CSV format and required multiple stages of preparation:

1. Imported raw CSV files into SQLite to create structured relational tables.
2. Performed data validation and aggregation using SQL, including joins across sessions, orders, products, and refunds.
3. Calculated key metrics such as total revenue, conversion rate, average order value, product-level sales, and refund metrics.
4. Exported aggregated query outputs into Python for further validation and formatting.
5. Cleaned and standardized the final datasets in Python and exported analysis-ready CSV files.
6. Imported the cleaned CSV files into Power BI to create the final interactive dashboard.

This approach ensured clean, accurate, and analysis-ready data before visualization.

KPIs (Key Performance Indicators):

1. **Total Revenue:** Total sales generated across all products
2. **Average Order Value (AOV):** $\text{Total Revenue} \div \text{Total Orders}$
3. **Refund Rate:** Percentage of refunded items relative to total items sold

FINDINGS:

1. The Original Mr. Fuzzy is the highest revenue-generating product and also leads in sales volume, making it the primary revenue driver.
2. The Forever Love Bear and The Birthday Sugar Panda contribute moderately to revenue but show different sales and refund patterns.
3. The Birthday Sugar Panda has the highest refund rate, indicating potential issues related to product quality, customer expectations, or fulfillment.
4. Higher-priced products tend to have lower refund rates, suggesting better perceived quality or customer satisfaction.
5. A small number of products contribute to the majority of revenue, highlighting a product concentration risk.

BUSINESS RECOMMENDATIONS:

1. Focus marketing and inventory planning around top-performing products such as The Original Mr. Fuzzy to maximize revenue.
2. Investigate the high refund rate for The Birthday Sugar Panda to identify quality, packaging, or description mismatches.
3. Consider revising pricing, product descriptions, or quality checks for lower-performing products to reduce refund-related losses.
4. Diversify the product portfolio to reduce dependency on a single high-revenue product.
5. Use refund rate metrics as an ongoing quality and customer satisfaction indicator.